

NALUT, Libyan Arab Jamahiriya

WMO: 620020

Lat: 31.867N

Lon: 10.983E

Elev: 621

StdP: 94.08

Time Zone: 2.00 (E02)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	5.8	-9.3	1.8	15.6	-7.7	2.1	15.5	14.5	10.0	12.4	9.2	5.1	270	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	40.0	20.3	38.4	20.0	36.7	19.7	24.4	33.6	23.2	31.6	22.2	30.5	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	18.1	27.9	20.9	16.8	26.6	19.9	15.7	26.0	77.4	33.9	72.2	31.6	68.3	30.2	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.9	8.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	11.3	12.1	15.9	19.2	23.4	27.1	29.3	29.2	27.0	22.2	16.2	12.2
	DBStd	7.46	2.27	3.23	4.07	4.31	3.93	4.21	3.53	2.77	3.33	3.58	3.62	3.23
	HDD10.0	38	11	13	2	0	0	0	0	0	0	0	1	11
	HDD18.3	829	219	178	100	41	4	1	0	0	0	8	86	193
	CDD10.0	3856	50	71	184	276	414	514	599	594	509	378	187	79
	CDD18.3	1604	0	2	24	66	161	264	341	336	259	127	22	3
	CDH23.3	18819	0	17	177	628	1652	3325	4626	4381	2953	961	91	9
CDH26.7	9939	0	4	48	246	748	1865	2761	2454	1489	310	14	0	
Wind	WSAvg	4.1	4.4	4.2	4.4	4.4	4.6	4.2	3.9	3.7	3.9	3.5	3.9	4.4
Precipitation	PrecAvg	123	11	14	29	16	2	2	1	0	8	17	18	18
	PrecMax	226	71	82	252	82	14	14	29	6	52	112	87	95
	PrecMin	37	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	55	12	17	43	23	4	4	5	1	12	25	23	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.4	25.6	30.4	34.3	37.2	41.6	42.6	40.3	38.9	33.2	28.7	24.6
		MCWB	9.6	12.2	12.8	14.9	18.3	21.0	21.7	20.2	20.8	18.0	16.0	12.1
	2%	DB	18.2	21.7	27.6	31.8	34.7	39.4	40.6	38.6	36.8	31.8	25.8	21.4
		MCWB	8.9	10.5	13.5	14.8	17.7	19.0	20.6	20.6	19.9	17.5	14.7	11.1
	5%	DB	16.5	19.4	24.9	29.6	32.9	37.2	38.6	37.0	35.2	30.1	23.9	19.0
		MCWB	8.4	10.2	12.9	14.4	17.2	19.1	20.4	20.0	19.5	17.2	13.8	10.4
	10%	DB	15.2	17.5	22.6	27.1	30.9	35.0	36.6	35.4	33.5	28.2	22.1	17.1
		MCWB	8.2	9.3	11.8	13.9	16.8	18.5	19.8	19.8	19.3	16.4	13.5	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	14.8	18.0	18.2	22.5	24.8	26.0	26.6	24.5	21.5	19.3	14.1
		MCDB	15.0	20.4	25.9	28.8	32.4	35.2	38.4	33.7	32.7	27.9	23.2	20.3
	2%	WB	10.7	12.5	15.6	16.5	20.1	22.6	23.8	24.5	23.0	20.2	17.3	12.3
		MCDB	14.6	19.6	22.9	26.4	30.4	33.2	33.5	32.7	29.7	27.3	22.6	18.4
	5%	WB	9.9	11.1	14.0	15.4	18.9	21.2	22.6	23.2	22.2	19.2	15.9	11.3
		MCDB	13.7	17.2	21.3	26.0	28.6	31.4	32.6	30.8	29.2	26.4	20.5	16.4
	10%	WB	9.2	10.1	12.7	14.4	18.0	20.0	21.4	22.1	21.3	18.2	14.8	10.5
		MCDB	13.1	15.6	20.2	24.8	27.4	29.6	31.2	29.6	28.5	24.5	19.8	15.2
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.7	8.8	10.1	11.1	11.4	11.1	10.3	9.8	8.2	7.2	6.7
		MCDBR	8.5	10.3	11.7	13.3	12.8	12.9	12.6	11.7	11.7	10.4	9.3	8.9
		MCWBR	4.0	5.4	5.1	4.6	4.9	4.5	5.0	4.4	4.6	4.3	4.2	4.7
	5% WB	MCDBR	6.5	9.4	9.9	11.5	11.7	11.9	12.3	11.1	11.0	8.8	7.3	7.6
MCWBR		3.8	5.8	5.2	4.8	5.1	5.2	5.5	5.7	5.0	4.5	4.3	4.6	
Clear-Sky Solar Irradiance	taub	0.343	0.368	0.455	0.510	0.533	0.489	0.486	0.479	0.502	0.454	0.387	0.350	
	taud	2.215	2.122	1.871	1.759	1.728	1.850	1.867	1.899	1.849	1.941	2.086	2.194	
	Ebn at Noon	864	877	820	792	776	808	808	808	767	772	799	830	
	Edh at Noon	115	139	193	225	235	207	202	193	195	165	130	112	
All-Sky Solar Radiation	RadAvg	3.53	4.42	5.57	6.58	7.23	7.81	7.88	7.20	6.00	4.76	3.72	3.18	
	RadStd	0.19	0.18	0.25	0.26	0.32	0.22	0.16	0.12	0.20	0.25	0.15	0.18	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page